

Supporting Information

Chemometric Diagnosis of Data-Splitting Effects in Molecular Property Prediction Benchmarks

S. Tong and Y. Wang

S1 Complete Paired Robustness Results

Table S1 reports the complete 20-seed paired results underlying the paired-effects analysis in the main manuscript. Positive reduction means that the target-balanced generalization gap was smaller than the ordinary scaffold gap. The p values are from two-sided Wilcoxon signed-rank tests followed by Holm correction across all 18 comparisons.

Table S1: Complete paired generalization-gap effects across 20 seeds.

Dataset	Model	Metric	Ordinary gap	Balanced gap	Mean reduction	Holm-adjusted p
BACE	Logistic regression	ROC-AUC	0.0952	0.0769	0.0183	0.0887
BACE	Random forest	ROC-AUC	0.0621	0.0738	-0.0117	0.1063
BACE	XGBoost	ROC-AUC	0.0659	0.0856	-0.0197	0.0292
BBBP	Logistic regression	ROC-AUC	0.1406	0.0565	0.0841	< 0.0001
BBBP	Random forest	ROC-AUC	0.1097	0.0761	0.0336	< 0.0001
BBBP	XGBoost	ROC-AUC	0.0984	0.0863	0.0121	0.1063
ClinTox	Logistic regression	ROC-AUC	0.1019	0.0472	0.0547	0.0028
ClinTox	Random forest	ROC-AUC	0.0818	0.0076	0.0741	< 0.0001
ClinTox	XGBoost	ROC-AUC	0.0066	0.0270	-0.0204	0.0292
HIV	Logistic regression	ROC-AUC	0.0362	0.1052	-0.0690	< 0.0001
HIV	Random forest	ROC-AUC	0.0450	0.1008	-0.0558	< 0.0001
HIV	XGBoost	ROC-AUC	0.0150	0.0923	-0.0774	< 0.0001
ESOL	Random forest	RMSE	1.0073	0.5047	0.5027	0.0332
ESOL	Ridge	RMSE	-0.0258	0.7225	-0.7484	0.0006
ESOL	XGBoost	RMSE	0.9607	0.3578	0.6029	< 0.0001
FreeSolv	Random forest	RMSE	0.9321	-0.1208	1.0529	< 0.0001
FreeSolv	Ridge	RMSE	0.6885	0.5980	0.0905	< 0.0001
FreeSolv	XGBoost	RMSE	1.1290	0.1807	0.9482	< 0.0001

S2 Model Ranking Stability

Split choice can change not only absolute performance estimates but also model rankings. Figure S1 summarizes the top-ranked model under each partitioning protocol. A change in the top-ranked model indicates that a leaderboard-style conclusion depends on the validation design.

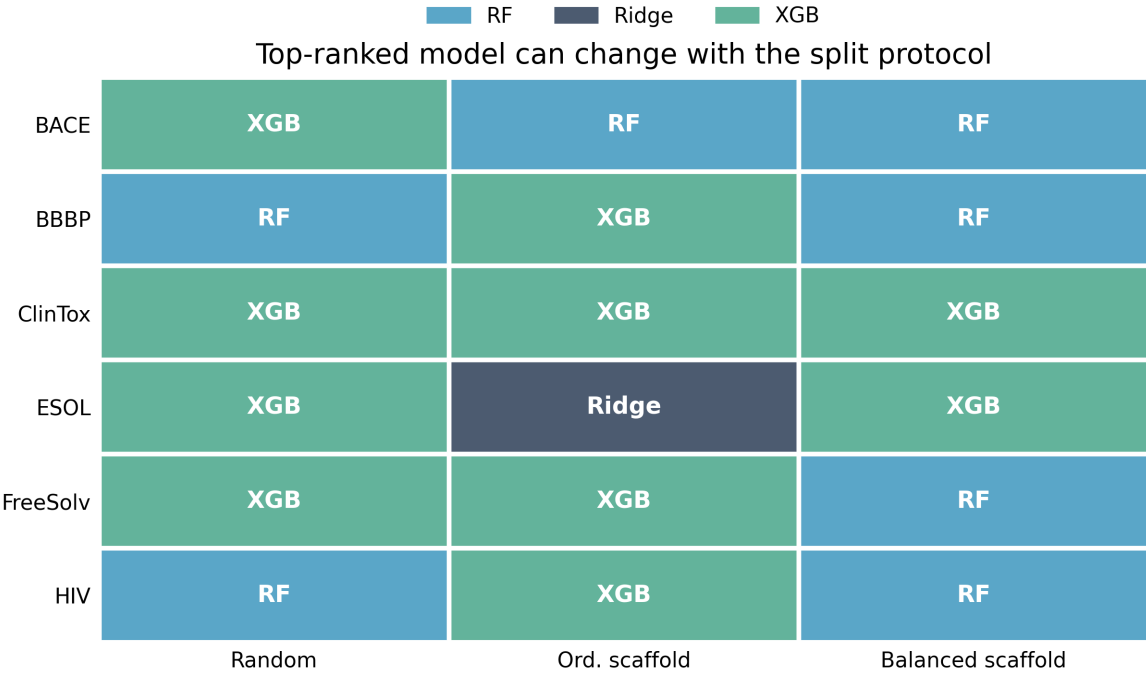


Figure S1: Top-ranked model by dataset and partitioning protocol. Changes across protocols show that comparative model conclusions can depend on test-set construction.

S3 Sensitivity Cases

Figure S2 retains cases in which target balancing changed the generalization gap strongly or changed its direction. These cases are diagnostically useful because they identify interactions between partition composition and model behaviour rather than a universal effect of balancing.

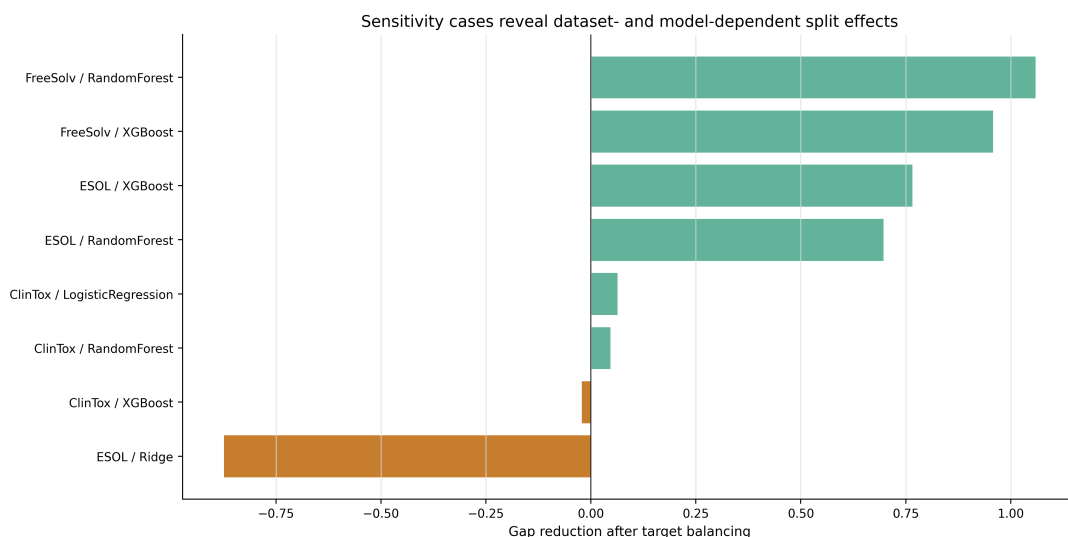


Figure S2: Sensitivity cases from the split audit. Positive values indicate reduction of the ordinary scaffold gap after target balancing; negative values indicate a larger target-balanced gap or a direction change.

S4 Reproducibility Notes

The complete workflow and machine-readable outputs are retained in the public repository. The 20-seed predictive analysis is implemented in `paper1_leakage_benchmark/scripts/robustness_20seeds.py`; the 5,000-draw random-scaffold audit is implemented in `paper1_leakage_benchmark/scripts/balanced_split_null_audit.py`; and the paired robustness figure is generated by `paper1_leakage_benchmark/scripts/plot_robustness20_figure.py`. The corresponding raw, paired, summary, diagnostic, null-assignment, and percentile tables are stored under `paper1_leakage_benchmark/results/tables/`. Fixed fingerprints, hyperparameters, seed values, and output names are encoded in the scripts so that the reported results can be regenerated without manual intervention.